

Coffee Sales Analysis - Project Report

Executive Summary

This project examines coffee retail sales performance by combining SQL Server for data querying and analysis with Power BI for visualization and reporting. The data is structured around a Fact table holding transactional measures such as sales, profit, cost of goods sold, marketing spend, inventory, and budget figures supported by two dimension tables, Product and Location, which describe what was sold and where.

The SQL layer of the project consists of 55 numbered queries, progressing from basic data retrieval and filtering through aggregation, multi-table joins, subqueries, CASE-based classification, set operations, window functions, and ROLLUP-based hierarchical summaries. On top of this, three views, three indexes, a stored procedure, and a user-defined function were built to keep the analysis reusable and reasonably efficient, along with transaction handling and basic error handling to demonstrate safe data modification practices. These SQL results were then carried into three Power BI dashboards: Regional & Market Performance, Product Performance, and a consolidated Coffee Sales Performance summary, which translate the underlying query outputs into an interactive format suitable for a business audience.

The analysis found that California leads the dataset in total sales, while Illinois shows a stronger profit margin, meaning the two states are strong on different measures rather than one being categorically ahead. Columbian emerged as the strongest overall product on both sales and profit, while Green Tea showed a negative profit margin worth reviewing further. Mint's inventory level and Caffè Mocha's marketing spend, relative to their respective sales positions, were also flagged as areas meriting closer attention. Together, the SQL and Power BI components of this project reflect a complete analytics workflow, from querying and validating raw data, to deriving findings and communicating them in a business-usable form.

Introduction

Coffee retail businesses accumulate transactional data across products, locations, and time, but that data only becomes useful once it is queried, organized, and interpreted with some rigor. This project works with a coffee sales dataset covering sales, profit, cost of goods sold, marketing expenditure, inventory levels, and budget figures across a number of U.S. states and coffee-related product types, and uses it to build out a full SQL-to-Power BI analytics workflow.

SQL was chosen as the foundation for this analysis because it is the standard way to extract precise, verifiable answers directly from a relational database. Working with SQL forces a certain discipline: before any number can be produced, the underlying question has to be phrased exactly, and the tables involved have to be joined correctly. Beyond basic SELECT statements, this project uses SQL Server and SSMS to build views for reusable reporting logic, indexes to support frequently used filtering and join patterns, a stored procedure and a user-defined function for repeatable product lookups, and transactions to safely test data-modifying operations before they would ever be applied for real.

Power BI was layered on top of this SQL work for a simple reason: query results, however accurate, are not something a business stakeholder typically wants to read row by row. Rebuilding the same underlying figures as an interactive dashboard with KPI cards, a geographic map, and filterable charts makes the findings usable by someone without any SQL background, while still tracing back to the same verified numbers produced by the SQL layer.

The two tools work together in a fairly linear way in this project. SQL is used first to validate the data, answer specific business questions, and establish what the numbers actually show. Power BI is then used to present those same findings visually, without introducing new calculations or new conclusions beyond what the SQL layer already established. This sequencing (analyze first in SQL, then visualize in Power BI) reflects a fairly standard analytics workflow and was intentional in how this project was structured.

Problem Statement

A coffee retail business generates data across products, states, and time, but without structured querying, that data does not answer any business question on its own. This project set out to use SQL to interrogate the dataset directly and understand how the business was actually performing across several dimensions: which products and locations were driving results, where spending was going, and whether that spending was aligned with outcomes.

Specifically, the analysis focused on the following core questions:

- Which products and product types generate the most sales and profit, and are any of them generating sales without generating profit?
- How does performance differ across states, both in terms of total sales and profit efficiency?
- How does marketing spend compare across products relative to the sales and profit those products actually produce?
- What do inventory levels look like by product, and do any of them appear misaligned with sales performance?
- Based on the available time-based data, is there any indication of change in sales between the two years covered?

The 55 SQL queries that make up the Master SQL file were written specifically to answer these questions, and the Power BI dashboards were built to present the resulting answers.

Project Objectives

- Analyse total sales, profit, and profit margin across the dataset
- Evaluate performance at the individual product and product-type level
- Examine inventory levels by product
- Assess marketing expenditure relative to sales and profit outcomes
- Compare performance across U.S. states, on both total sales and profit efficiency
- Identify products that consistently perform well or require closer review, based strictly on the data
- Apply core SQL Server techniques, joins, aggregation, subqueries, window functions, ROLLUP, views, a stored procedure, a user-defined function, transactions, and error handling to answer the above questions directly against the data
- Translate the SQL-derived findings into interactive Power BI dashboards for business use

Dataset Description

The project is built on a database named Project, structured around one fact table and two dimension tables that together capture the coffee retail business's transactional data.

Table	Key Fields	Purpose
Product	ProductID, Product, Product_Type	Describes what is being sold the product name and its broader category (e.g., Coffee, Espresso, Herbal Tea, Tea, Regular)
Location	Area_Code, State	Describes where a transaction took place
Fact	Date, ProductID, Area_Code, Sales, Profit, COGS, Marketing, Total_Expenses, Inventory, Budget_Margin	The transactional table holding all measurable business figures; every other table connects back to this one

The relationships between the tables are straightforward: Fact.ProductID links to Product.ProductID, and Fact.Area_Code links to Location.Area_Code. No additional tables or relationships exist in the database beyond these three.

Tools and Technologies

Tool / Technology	Purpose in the Project
Microsoft SQL Server / SSMS	Database engine used to host the Product, Location, and Fact tables and execute all 55 project queries
T-SQL	Query language used for filtering, aggregation, joins, subqueries, CASE logic, set operations, window functions, ROLLUP, views, indexes, a stored procedure, a user-defined function, transactions, and error handling
Power BI	Used to build three interactive dashboards on top of the SQL-derived data, incorporating KPI cards, a map visual, charts, and slicers

Database Design and Data Model

The database follows a fact-dimension structure, a common and practical way to organize transactional business data for analysis. The Fact table stores the measurable numbers sales, profit, COGS, marketing, expenses, inventory, and budget margin while the Product and Location tables hold the descriptive attributes that give those numbers context.

This separation matters because it keeps repeated descriptive text (product names, state names) out of the transactional table, and it keeps joins simple: any query that needs to know "which product" or "which state" a row belongs to just reference the relevant dimension table through its key, rather than duplicating that information across every row of the Fact table. It's worth noting that this project doesn't claim the model has been formally normalized to a specific normal form (that wasn't something tested or established here), but the fact-dimension split itself is a sound and standard structure for this kind of analysis.

Relationship	Description
Fact.ProductID → Product.ProductID	Links each transaction to its product name and product type
Fact.Area_Code → Location.Area_Code	Links each transaction to the state it occurred in

SQL Analysis and Implementation

The SQL work in this project was approached as a progression, starting with basic data validation, moving through filtering and aggregation to answer specific numeric questions, and building up to more advanced database objects that made the analysis reusable and safer to extend. The Master SQL file contains 55 numbered queries in total, along

with the supporting views, indexes, stored procedure, function, transaction demonstrations, and error handling described below. Rather than listing every individual query, they are organized here by technique, with a short representative example included where it adds clarity.

Basic Retrieval and Filtering: The project begins with simple SELECT statements used to confirm that the Product, Location, and Fact tables had imported correctly, followed by filtering queries, for example isolating transactions where expenses crossed a certain threshold, and basic sorting, such as listing states alphabetically.

```
SELECT COUNT(DISTINCT State) AS No_of_states FROM Location;
```

Aggregation and Grouping: SUM, AVG, MIN, and MAX functions were used throughout to answer direct numeric questions, such as total marketing spend on a specific product, average sales within a given area code, and average inventory per product, among others.

```
SELECT ProductID, AVG(Inventory) AS Avg_Inventory
FROM Fact
GROUP BY ProductID;
```

Joins: Since the Fact table on its own doesn't carry product names or state names, joins to Product and Location were essential for nearly every business-facing question in this project, for instance calculating profit by state, or sales and profit broken down by both state and product together.

```
SELECT L.State,
       SUM(F.Profit) AS TotalProfit,
       SUM(F.Sales) AS TotalSales
FROM Fact F
JOIN Location L ON F.Area_Code = L.Area_Code
GROUP BY L.State;
```

Subqueries: A handful of queries used a nested SELECT to compare individual rows against a separately calculated aggregate, for example identifying products with sales above the dataset's overall average, or finding the second-highest sales value in the table.

```
SELECT *
FROM Fact
WHERE Sales > (SELECT AVG(Sales) FROM Fact);
```

CASE Statements: Conditional logic was applied directly within queries to classify rows without needing a separate processing step, such as labelling transactions as Profit or Loss based on an expense threshold, and grouping sales into High, Medium, and Low bands.

```
SELECT ProductID, Sales,
       CASE
         WHEN Sales >= 500 THEN 'High'
         WHEN Sales >= 200 THEN 'Medium'
         ELSE 'Low'
       END AS SalesCategory
FROM Fact;
```

Set Operations: UNION and INTERSECT were applied to Area_Code values to compare what exists in the Fact table against what exists in the Location table, which is a useful sanity check for data consistency between the two.

Window Functions: DENSE_RANK() was used to rank sales without leaving gaps in the ranking sequence, and SUM() OVER() was used to build a running total of sales ordered by date, both of which allow row-level detail and aggregate context to sit side by side in the same result set.

```
SELECT Date, Sales,  
       SUM(Sales) OVER (ORDER BY Date) AS Running_Total_Sales  
FROM Fact;
```

ROLLUP: A hierarchical aggregation of total sales by Date and ProductID was built using ROLLUP, which adds subtotal and grand-total rows on top of the normal grouped output, useful for seeing both the detail and the summary in a single query.

Transactions: BEGIN TRANSACTION paired with ROLLBACK was demonstrated twice: once for an UPDATE and once for a DELETE, so that both operations could be tested against the real table structure without any risk of permanently altering the underlying data.

Views: Three views were created to package up commonly needed joins so the same logic wouldn't need to be rewritten repeatedly across the project:

- vw_SalesReport: combines Date, Product, Product_Type, State, Sales, Profit, COGS, and Marketing into a single reporting view
- vw_StatePerformance: aggregates total sales and profit by state
- vw_ProductPerformance: aggregates total sales, profit, marketing, and average inventory by product

Indexes: Three indexes were added to the Fact table (IX_ProductID, IX_Area_Code, and IX_Date) on the columns most repeatedly used across the project's joins and filters. These are intended to improve query efficiency for those recurring access patterns; no formal benchmarking was carried out to measure the actual performance difference.

Stored Procedure: dbo.Get_Product_By_Type accepts a @ProductType parameter and returns matching products, replacing what would otherwise be a manually repeated WHERE clause each time a lookup was needed.

```
EXEC dbo.Get_Product_By_Type 'Coffee';
```

User-Defined Function: dbo.GetProductType performs the same underlying lookup as a table-valued function, which allows it to be used inline as part of a larger query for instance, inside a JOIN rather than only as a standalone execution like the stored procedure.

Error Handling: A TRY/CATCH block was used to wrap a basic SELECT statement, demonstrating that an error can be caught and a readable message returned through ERROR_MESSAGE () rather than the query simply failing without explanation.

Business Questions and Findings

1 State Performance: Sales versus Profit Efficiency

One of the first questions this project set out to answer was which state was actually performing best and whether "best" meant the same thing depending on how it was measured. By joining Fact to Location on Area_Code, grouping by state, and aggregating sales, profit, and profit margin (via Query 17 and the vw_StatePerformance view), the analysis found that California recorded the highest total sales of any state, while Illinois recorded a higher profit margin than California.

This distinction matters. California is clearly the leading market by overall business scale, since the volume of sales there outpaces every other state in the dataset. But Illinois converts its sales into profit more efficiently. Rather than naming a single "best" state, the more accurate reading is that California and Illinois are strong on two different measures, and both are worth understanding on their own terms rather than collapsing into one ranking.

2 Product Performance: Identifying the Strongest Overall Product

To determine which product performed best overall, Fact was joined to Product and grouped by product name, with rankings computed separately by total sales and total profit (Queries 30, 31, 40, and 45). Columbian recorded the highest total sales and the highest total profit of any product in the dataset, a fairly clear result since it's unusual for the same product to lead on both measures simultaneously.

This makes Columbian the strongest overall product performer in the dataset by a reasonably wide margin, and a natural anchor point for any product-level comparison elsewhere in the analysis.

3 Product Profitability: Sales Without Profit

Not every product that sells well is actually profitable, which is why the analysis also filtered specifically for products with zero or negative profit (Query 48). This surfaced Green Tea, which shows a negative profit margin despite generating measurable sales volume.

The finding itself is clear from the data, but the reason behind it is not. It would be a mistake to assume the cause is pricing, cost structure, or any other single factor without further investigation. The dataset simply shows that Green Tea's economics don't currently work in the business's favor, and that this requires a closer look at its pricing, costs, or broader product economics before any conclusion is drawn.

4 Marketing: Spend Relative to Outcome

Grouping Fact by product and aggregating marketing spend alongside profit (Query 37) showed that Caffè Mocha receives the highest marketing investment of any product, yet ranks below Columbian in both sales and profit.

This is worth flagging as a mismatch, but it should not be read as proof that Caffè Mocha's marketing is failing or that its budget should simply be redirected. The data shows an association between spend and outcome, but nothing here establishes that reducing or reallocating that spend would actually change the result.

5 Inventory: Levels Relative to Demand

Averaging inventory levels by product (Queries 10, 36, and 46) showed that Mint carries the highest average inventory of any product in the dataset, despite not being among the top sellers.

This gap between inventory held and product performance is a reasonable signal to investigate. Whether the inventory level is genuinely excessive, seasonal, or tied to some other operational factor isn't something the current data can answer on its own, so it's best treated as an open question rather than a confirmed inventory problem.

6 Time Trend: Early Signs of Growth

Finally, to see whether sales had changed over time, Fact was grouped by year and month and summed (Queries 33 and 55). Based specifically on the available January observations, sales increased by 4.36% between January 2010 and January 2011.

It's important to be precise about what this does and doesn't show. This is a single-month comparison, not a full seasonal or annual trend line across both years, so it's best treated as an early positive signal rather than confirmation of sustained growth across the whole period.

Key Findings and Business Insights

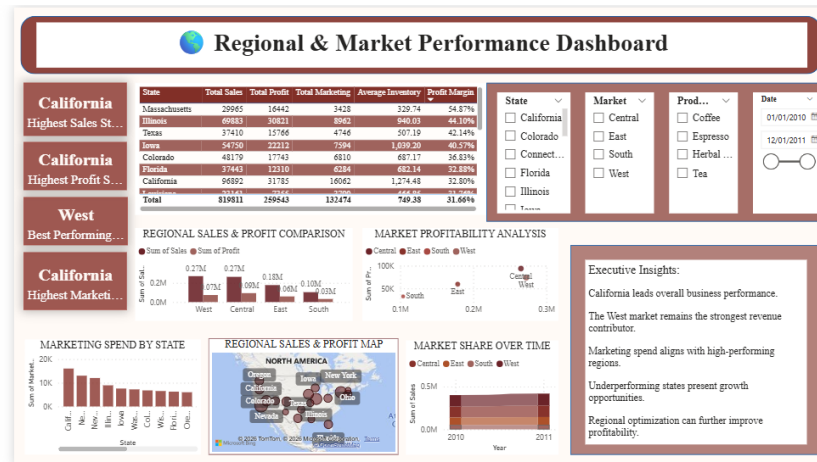
Area	Finding	Business Significance
Sales	Total sales across the dataset came to approximately 819,811 units	Establishes the overall scale of the business captured in the dataset
Profitability	Overall profit margin came out to 31.66%	A healthy baseline figure, though not evenly distributed across products or states
Geographic Performance	California leads on total sales; Illinois has a stronger profit margin	Sales scale and profit efficiency are led by two different states
Product Performance	Columbian leads on both sales and profit; Regular Espresso records the lowest sales	Columbian anchors overall performance; Regular Espresso's low sales don't necessarily mean it's unprofitable
Inventory	Mint holds the highest average inventory of any product	Worth reviewing against actual demand, though not confirmed as overstocked
Marketing	Caffe Mocha receives the highest marketing investment of any product	Doesn't correspond to a top sales or profit position an association worth reviewing
Time Trend	Sales rose 4.36% from January 2010 to January 2011	Based on available January data only, not a full annual comparison

Taken together, these findings paint a picture of a business that performs solidly overall. A 31.66% margin and steady early sales growth are both reasonable indicators of health, but there are clear pockets that don't match the rest of the pattern. California's scale doesn't automatically mean it's the most efficient state; Caffe Mocha's marketing investment doesn't automatically mean it's the top seller; and Mint's inventory doesn't automatically mean it's moving fast. None of these gaps are large enough to call the business troubled, but each is specific enough to be worth a closer, targeted look rather than a blanket conclusion.

Power BI Dashboards

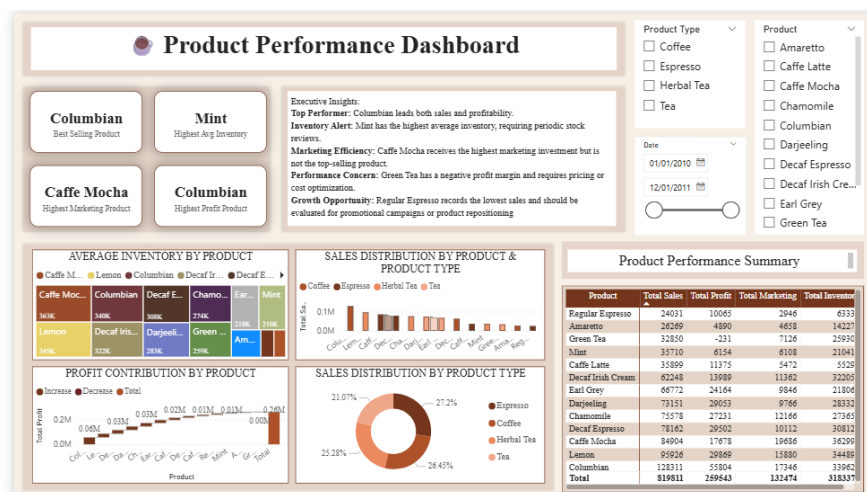
The SQL analysis was carried into three Power BI dashboards, each built to support a different layer of the business questions this project set out to answer.

Regional & Market Performance Dashboard



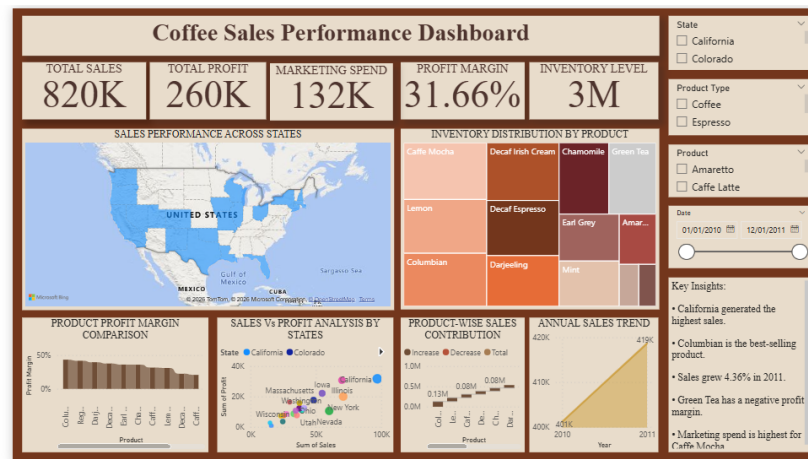
This dashboard exists to answer the geographic side of the analysis: how sales, profit, and marketing spend vary by state and by market region (Central, East, South, West). It includes callout cards for the top state by sales, the top state by profit, the best-performing market, and the top state by marketing spend, alongside a sortable state performance table covering Total Sales, Total Profit, Total Marketing, Average Inventory, and Profit Margin. Supporting visuals include a regional sales-vs-profit comparison chart, a market profitability scatter plot, a marketing spend by state chart, a regional sales and profit map, a market share over time chart, and a written executive insights panel. The page can be filtered by State, Market, Product Type, and Date.

Product Performance Dashboard



This dashboard focuses on individual product-level performance, supporting the product and inventory questions from the analysis. It includes callout cards for the best-selling product, the product with the highest average inventory, the product with the highest marketing spend, and the highest-profit product, along with a product performance summary table covering Total Sales, Total Profit, Total Marketing, and Total Inventory. Supporting visuals include an average inventory by product chart, a sales distribution by product and product type chart, a profit contribution chart by product, and a sales distribution by product type chart. The page can be filtered by Product Type, Product, and Date.

Coffee Sales Performance Dashboard



The third dashboard consolidates geography and product into a single executive-level view. It leads with KPI cards for Total Sales, Total Profit, Marketing Spend, Profit Margin, and Inventory Level, followed by a state-level sales map, an inventory distribution chart by product, a profit margin comparison chart across products, a sales-vs-profit chart by state, a product-wise sales contribution chart, an annual sales trend chart, and a written key insights panel. The page can be filtered by State, Product Type, Product, and Date.

Dashboard Insights

Power BI, in this project, functions as a visualization and communication layer built directly on top of the SQL analysis. It doesn't introduce any new calculations or new findings beyond what the SQL layer already established. Its value is in making those same findings accessible and interactive for someone without a SQL background. That relationship is visible across all three dashboards. California's lead in total sales and Illinois's stronger profit margin both appear consistently in the state performance table and executive insights panel of the Regional & Market Performance dashboard, matching exactly what the underlying SQL query returned. Columbian's position as the top product on both sales and profit, and Green Tea's negative profit margin, are similarly consistent across the Product Performance and summary dashboards: the same numbers, just presented differently depending on which page is being viewed.

The marketing and inventory findings follow the same pattern. Caffe Mocha's marketing spend relative to its sales rank is called out directly in the Product Performance dashboard's insights panel, and the January-to-January sales increase of 4.36% is reflected in the annual sales trend visual on the summary dashboard. Because every one of these figures traces back to the same SQL queries described in Section 10 and Section 11, there's no discrepancy between what the dashboards show and what the underlying analysis found; the dashboards are simply a different way of viewing the same verified results.

Business Recommendations

1. Product Strategy: Colombian's consistent lead in both sales and profit makes it a reasonable candidate for continued prioritization in stocking and promotion. For Green Tea, the negative profit margin is worth investigating further: a review of its pricing, cost structure, or broader product economics would help clarify what's actually driving the shortfall before any specific action is taken.
2. Inventory Management: Mint's inventory level stands out as the highest in the dataset without a corresponding lead in sales. This is worth reviewing against actual demand and sales velocity to understand whether current stocking levels are appropriate, rather than assuming overstocking as a given.
3. Marketing Effectiveness: Caffe Mocha's marketing spend doesn't currently align with its sales or profit rank relative to Colombian. Before reallocating any budget, it would be worth reviewing campaign-level returns for both products to understand what's actually driving the mismatch. The current data shows an association, not a cause.
4. Geographic Focus: Since California and Illinois are strong on different measures, it may be worth examining what's driving Illinois's stronger profit margin specifically, to see whether any of that efficiency could reasonably be applied to California's larger sales base.
5. Sales Opportunities: Regular Espresso's low sales position is worth a closer look, but this shouldn't be treated as evidence that the product is unprofitable or a candidate for discontinuation that hasn't been established by the current analysis.
6. Profitability Monitoring: Given that both the Illinois and Green Tea findings show sales volume alone doesn't tell the full profitability story, it would be worth continuing to monitor profit margin at both the product and state level as a routine check going forward.

SQL Performance and Database Design Considerations

The three indexes on the Fact table (IX_ProductID, IX_Area_Code, and IX_Date) were added because these are the columns most repeatedly used across the project's joins and filtering conditions. The intent behind them is to improve query efficiency for these recurring access patterns; it's worth being clear, though, that no formal benchmarking was carried out to measure any actual before-and-after performance difference, so any efficiency gain here is expected rather than demonstrated.

The three views built for this project (vw_SalesReport, vw_StatePerformance, and vw_ProductPerformance) exist to avoid rewriting the same join logic every time a particular breakdown is needed. This also keeps the SQL layer and the Power BI dashboards consistent with each other, since both can pull from the same underlying view definition rather than each recreating the logic independently.

The stored procedure and the user-defined function both perform the same underlying lookup (products filtered by type), but they're built for different situations. The stored procedure, dbo.Get_Product_By_Type, is meant to be run as a standalone call via EXEC. The function, dbo.GetProductType, is designed to be used inline as part of a larger query, such as inside a join, where a stored procedure couldn't be used directly.

Finally, the transaction demonstrations, an UPDATE and a DELETE, each wrapped in BEGIN TRANSACTION and followed by ROLLBACK, exist to show that both operations can be tested safely against the live table structure without any risk of permanently altering the underlying data. This is standard practice before running any write operation for real, and was included here specifically to demonstrate that awareness.

Challenges and Learning Outcomes

The biggest recurring challenge in this project was translating a plain-language business question into a correct SQL query. It sounds simple in principle, but getting something like the state-wise profit-and-sales query right meant being precise about exactly which table each column came from and making sure the join condition was exact. A small mistake there doesn't throw an error, it just quietly changes the result, which makes it easy to miss if you're not double-checking the output against what you'd expect.

ROLLUP and the window functions took a bit more work to get comfortable with than the more straightforward aggregation queries. Their output structure (the extra subtotal and grand-total rows that ROLLUP adds, or the additional ranking column that DENSE_RANK() introduces) isn't something that's obvious until the query is actually run and compared side by side with a plain GROUP BY result.

Beyond the purely technical side, this project also pushed me to be more careful about the difference between what a query actually proves and what I might be tempted to conclude from it. The Caffè Mocha marketing figures are a good example: it would have been easy to write that the marketing "isn't working," but the data only shows a mismatch between spend and sales rank, not that the spend is the reason for anything. Keeping that distinction consistent across the whole report took more discipline than I expected going in.

On the Power BI side, the main learning curve was less about the tool itself and more about deciding which of the 55 SQL queries were actually worth turning into a visual. Not every result needed a chart, and figuring out which findings genuinely mattered to a business audience versus which were just useful intermediate steps in the SQL analysis was its own kind of judgment call.

Limitations

- The time-based comparison in this project is based on available January 2010 and January 2011 observations rather than a complete monthly or seasonal time series across both years, which limits how confidently any broader trend can be claimed.
- The dataset is U.S.-specific, so the findings don't necessarily generalize to other markets.
- No customer-level data is included, so the analysis is limited to the product, location, and time dimensions available in the Fact table.
- No external market or competitor data is available, meaning all findings reflect only what's inside the Product, Location, and Fact tables.
- The relationship between marketing spend and sales or profit outcomes is observational: the project identifies an association, not a causal relationship.

Future Scope

- Incorporate more complete historical data to properly assess seasonality and longer-term trends beyond the available January snapshots
- Apply forecasting techniques to project future sales or profit by product and state
- Bring in customer-level data, if it becomes available, to support segmentation analysis
- Conduct campaign-level marketing analysis to better understand the Caffè Mocha spend-versus-sales mismatch
- Build a more detailed inventory forecasting approach informed by the Mint inventory finding
- Automate the Power BI dashboards against a live SQL Server connection rather than a static extract

Conclusion

This project took a coffee sales dataset through a complete analytics workflow, starting with validating the imported data and running 55 structured SQL queries across a wide range of techniques (basic retrieval and filtering, aggregation, joins, subqueries, CASE logic, set operations, window functions, ROLLUP, transactions, and error handling), supported by three views, three indexes, a stored procedure, and a user-defined function. Each of these was built to answer a specific, defined business question rather than as an exercise in isolation. The findings that came out of this SQL work were substantive and, in a few places, more nuanced than a single headline number would suggest. California leads on total sales, but Illinois converts sales into profit more efficiently. Columbian stands out clearly as the strongest overall product. Green Tea, Mint, and Caffè Mocha each surfaced a specific area, profitability, inventory, and marketing spend respectively, that doesn't match the rest of the business's pattern and is worth a closer, targeted look.

Power BI's role in this project was to take these SQL-verified findings and make them accessible to someone without a database background, through three dashboards that stayed faithful to the underlying query results rather than introducing anything new. Throughout the analysis, a deliberate effort was made to separate what the data actually proves from what might be tempting to conclude from it, particularly around marketing effectiveness and inventory levels, where an observed mismatch doesn't automatically mean a confirmed problem.

Ultimately, this project reflects both the technical and analytical sides of business analytics: writing correct, efficient SQL to extract real answers from a relational database, and then interpreting those answers with the kind of caution that keeps a report useful rather than overstated. That combination, solid technical execution paired with careful, appropriately bounded business interpretation, is the primary value this project set out to demonstrate.